

Yap Hong Ping

Singapore PR | +65-97773450 | yhp2130@gmail.com | [linkedin.com/in/yaphongping](https://www.linkedin.com/in/yaphongping) | yhp2130.github.io
Agentic AI · RAG · NLP · Computer Vision · MLOps

1 APPLIED AI ENGINEER | DATA SCIENTIST | 8+ YEARS PRODUCTION AI

Applied AI Engineer and Data Scientist with 8+ years delivering end-to-end production AI systems — Agentic RAG pipelines, LLM fine-tuning (BART, RoBERTa), computer vision, and full-stack deployment (FastAPI, React.js, OpenShift CI/CD). Delivered measurable outcomes across 5 shipped systems: 96% reduction in knowledge-retrieval time (2hr → 5min), F1=0.8 NLP clause extraction (200 contracts), 70% CV accuracy (receipt OCR), 100+ invoices automated/month, and concept-to-production MVPs in under 2 weeks using GitHub Copilot SpecKit. Open to Applied AI Engineer and Data Scientist roles in software, fintech, and deep-tech industries

2 KEY STRENGTHS

- Data Science: Frames ambiguous problems into measurable AI objectives; iterates via experimentation, RAGAS evaluation, and error analysis
- AI Application: Delivers ML, Deep Learning (CV & NLP), and Generative AI (LLM/RAG) with user requirements, feedback, and adoption in scope
- Full-Stack Deployment: Integrates models into web applications (React.js, Vue.js, FastAPI) meeting enterprise architecture standards
- Stakeholder Engagement: Agile delivery with regular demos, retrospectives, and change management driving measurable production adoption

3 CORE SKILLS · PYTHON

- ML / Deep Learning: PyTorch, TensorFlow, Scikit-learn, HuggingFace (BART, RoBERTa, Nougat, DETR, LayoutLM)
- Agentic AI & RAG: LangChain (ReAct, tool-calling), OpenAI AsyncOpenAI, Elasticsearch kNN (HNSW), Agentic Chunking, Prompt Engineering
- Backend & Full-Stack: FastAPI, FastMCP, React.js (TypeScript/SSE), Vue.js, Streamlit, Docker, Kubernetes, OpenShift
- MLOps: GitLab CI/CD, ArgoCD, Helm, Artifactory, OpenShift (dev/staging/prod) · Data: Elasticsearch, Oracle/MySQL/SQLite, Logstash
- AI Platforms: Datarobot, Dataiku, Dify, SAP Joule, Copilot Studio, GitHub Copilot

4 PROFESSIONAL EXPERIENCE

4.1 INFINEON TECHNOLOGIES ASIA PACIFIC — STAFF ANALYST, DATA SCIENTIST (NOV 2021–PRESENT)

Agentic Process Knowledge RAG

- Delivered production-ready multi-agent virtual consultant in <2 weeks (GitHub Copilot SpecKit); custom async ReAct — Planner → Phase Sub-Agents (5 specialisations) → Checker; ExecutionPlan/SubGoal dependency graph; 4 knowledge sources (ARIS, Confluence, Intranet, PRIMA)
- React.js + TypeScript frontend with SSE streaming; per-message thumbs-up/down feedback (SQLite); GitLab CI/CD → ArgoCD → Helm → OpenShift (dev/staging/prod) + Artifactory; per-agent YAML config
- **Stack:** Python · OpenAI AsyncOpenAI · FastAPI · React.js · GitLab CI/CD · ArgoCD · Helm · OpenShift · Artifactory

Document Retrieval & Agentic RAG Chatbot

- 3-generation evolution across ~1,000 documents (TF-IDF+SBERT → Elasticsearch kNN → LangChain ReAct agent); 96% query time reduction (2hr → 5min); 5 chunking strategies evaluated — agentic chosen; SFT-Mistral embedder via RAGAS; Docling + EasyOCR ingestion
- 3 role-gated LangChain tools (abbreviation, kNN search, BART contract clauses via LLM-to-SQL); ~100 queries/day; QnA feedback loop for continuous improvement

- **Stack:** Python · LangChain · Elasticsearch (kNN HNSW) · SFT-Mistral · Docling · FastAPI · React.js · Nginx · OpenShift

NLP Contract Clause Extraction

- Fine-tuned RoBERTa-large-MNLI (8-class logistics classifier, F1=0.8) + BART seq2seq on 200 labelled contracts; Nougat PDF transcription + DETR table detection preprocessing; human-in-the-loop Vue.js review; wrapped as LangChain agent tool inside ClaireGPT
- **Stack:** Python · HuggingFace (RoBERTa-MNLI, BART) · PyTorch · Nougat · DETR · FastAPI · Vue.js · GitLab CI/CD · Oracle DB

Travel Expense Receipt Intelligence

- SqueezeNet CNN (7 doc classes) + Faster RCNN segmentation + EasyOCR (custom weights) + regex NER; 7-flag rule validator; event-driven SAP Concur v4 webhook replacing batch; 70% accuracy on test samples
- **Stack:** Python · PyTorch (SqueezeNet · Faster RCNN) · EasyOCR · FastAPI · SAP Concur v4 API · SAP BTP

Distributor Invoice Audit Automation

- Configurable rule engine for AP invoice reconciliation (100+ invoices/month): 6 match flags (SMD, qty, price, currency, SAP#, date); fuzzy matching (thefuzz) + LLM fallback for CJK company names; ABBYY Vantage multilingual extraction (Chinese, German, English); Vue.js exception dashboard
- **Stack:** Python · ABBYY Vantage · FastAPI · Vue.js · Elasticsearch · Logstash · Docker
- **Team & Delivery:** Agile 2-week sprints, stakeholder demos and retrospectives; intern mentoring with structured onboarding and task reviews

4.2 PSA CORPORATION — PRINCIPAL ENGINEER, SMART SYSTEMS SOLUTIONS (JUL 2018–NOV 2021)

- Predictive Maintenance (Maritime Equipment): Semi-supervised pipeline (anomaly detection → supervised classification) on sensor/log data from >20 machines at 0.5s sampling; ~8 man-hours/week saved via continuous monitoring dashboard · Stack: Python · Scikit-learn
- CV Equipment Monitoring (POC): Object detection for structural crack and roller wear defect detection; custom training with image augmentation · Stack: Python · PyTorch
- Cloudera Hadoop data pipelines for large-scale industrial sensor ingestion and feature engineering; integrated with real-time monitoring dashboards · Stack: Python · Hadoop · Spark

4.3 PSA CORPORATION — SENIOR MECHANICAL ENGINEER (JUL 2013–JUL 2018)

- Automated component life tracking and failure pattern analysis; translated mechanical domain expertise into predictive monitoring criteria applied in subsequent ML system design

5 EDUCATION

- Master of Technology (Knowledge Engineering), National University of Singapore – Institute of Systems Science (NUS-ISS) | 2018–2020
- Bachelor of Engineering (Mechatronic Engineering), Nanyang Technological University (NTU) | 2009–2013